MATH73235 Spring 2025 Quiz 8

First name:
Last name:
Student ID:

*Name and Student ID must be filled in for this page to be graded.*

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*Preamble:* You may use any result from the "Helpful Results" section.

**Problem 1.** *Use Fourier Series to obtain a standing wave solution $u(x,t)$ to the 1D wave-equation for a string of length $L = 1$ and $c^2 = 1$, when the initial velocity is zero, and the initial deflection is given by $f(x) = 0.01\left(\sin(\pi x) - \frac{1}{2}\sin(2\pi x)\right)$.*

***[5 marks]***

*Solution.*

$\square$

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*Preamble:* You may use any result from the "Helpful Results" section.

**Problem 2.** *Write down the D'Alembert's solution $u(x,t)$ to the 1D wave-equation with initial conditions $u(x,0) = f(x)$ and $u_t(x,0) = 0$ when*

$$f(x) = 0.01x(1-x).$$

***[5 marks]***

*Solution.*

$\square$

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Extra Page: Must submit even if unused.

Helpful Results

- The one-dimensional wave equation $u_{tt} = c^2 u_{xx}$ with standing wave boundary conditions $u(0,t) = 0 = u(L,t)$ for $t \geq 0$ and $L > 0$, and initial conditions $u(x,0) = f(x)$, $u_t(x,0) = 0$ for $0 \leq x \leq L$, has solution based on Fourier Series given by

$$u(x,t) = \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi c}{L} t\right) \sin\left(\frac{n\pi}{L} x\right),$$

  with $A_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi}{L} x\right)\, dx$.

- The one-dimensional wave equation $u_{tt} = c^2 u_{xx}$ with initial conditions $u(x,0) = f(x)$, $u_t(x,0) = 0$ for $-\infty < x < \infty$, and no specific boundary conditions, has solution based on D'Alembert's work:

$$u(x,t) = \frac{1}{2}\left[f(x-ct) + f(x+ct)\right].$$